

Vector Algebra

▢ Vector: Vector is a quantity having both magnitude and direction.

Example: displacement, velocity, force.

▢ Scalar: Scalar is a quantity having only magnitude.

Example: Mass, length, time

▢ Magnitude of a vector: $|\vec{A}| = \sqrt{A_1^2 + A_2^2 + A_3^2}$

IF $\vec{A} = 2\hat{i} + 3\hat{j} - 5\hat{k}$ then

$$|\vec{A}| = \sqrt{2^2 + 3^2 + (-5)^2}$$
$$= \sqrt{38}$$

▢ Vector addition and subtraction:

▢ IF $\vec{A} = 2\hat{i} + 3\hat{j} - 5\hat{k}$, $\vec{B} = 3\hat{i} + 2\hat{j} + 2\hat{k}$ then find $\vec{A} + \vec{B}$ and $\vec{A} - \vec{B}$.

Solution: $\vec{A} + \vec{B} = (2\hat{i} + 3\hat{j} - 5\hat{k}) + (3\hat{i} + 2\hat{j} + 2\hat{k})$

$$= 5\hat{i} + 5\hat{j} - 3\hat{k}$$

$$\vec{A} - \vec{B} = (2\hat{i} + 3\hat{j} - 5\hat{k}) - (3\hat{i} + 2\hat{j} + 2\hat{k})$$
$$= -\hat{i} + \hat{j} - 7\hat{k}$$

▢ IF $\vec{A} = 3\hat{i} - 2\hat{j} + \hat{k}$, $\vec{B} = 2\hat{i} - 4\hat{j} - 5\hat{k}$, $\vec{C} = -2\hat{i} + \hat{j} - 3\hat{k}$ then find the magnitude of

① $\vec{A} + \vec{B} + \vec{C}$

$$(ii) 2\bar{A} - 4\bar{B} - 3\bar{C}$$

Solution:

$$(i) \bar{A} + \bar{B} + \bar{C} = 3\bar{i} - 5\bar{j} - 7\bar{k}$$

$$\therefore |\bar{A} + \bar{B} + \bar{C}| = \sqrt{3^2 + (-5)^2 + (-7)^2} \\ = \sqrt{83}$$

Dot product: The dot product of two vectors $\bar{A}$ and $\bar{B}$ denoted by $\bar{A} \cdot \bar{B}$ is defined as the product of the magnitude of $\bar{A}$ and $\bar{B}$ and the cosine of the angle θ between them.

$$\bar{A} \cdot \bar{B} = |\bar{A}| |\bar{B}| \cos \theta$$

Cross product: The cross product of two vectors $\bar{A}$ and $\bar{B}$ denoted by $\bar{A} \times \bar{B}$ is defined as the product of the magnitude of $\bar{A}$ and $\bar{B}$ and the sine of the angle θ between them.

$$\bar{A} \times \bar{B} = |\bar{A}| |\bar{B}| \sin \theta$$

Q1: find the angle between $\bar{A} = 2\bar{i} + 2\bar{j} - \bar{k}$ and $\bar{B} = 6\bar{i} - 3\bar{j} + 2\bar{k}$.

Solution: we know,

$$\bar{A} \cdot \bar{B} = |\bar{A}| |\bar{B}| \cos \theta \text{ ——— (i)}$$

∴ Again,

$$\begin{aligned}\vec{A} \cdot \vec{B} &= (2\hat{i} + 2\hat{j} - \hat{k}) \cdot (6\hat{i} - 3\hat{j} + 2\hat{k}) \\ &= 12 - 6 - 2 \\ &= 4\end{aligned}$$

$$\begin{aligned}\therefore |\vec{A}| &= \sqrt{2^2 + 2^2 + 1^2} \\ &= \sqrt{9} \\ &= 3\end{aligned}$$

$$\begin{aligned}|\vec{B}| &= \sqrt{6^2 + (-3)^2 + 2^2} \\ &= \sqrt{36 + 9 + 4} \\ &= 7\end{aligned}$$

∴ from (i) we get,

$$\begin{aligned}\vec{A} \cdot \vec{B} &= |\vec{A}| |\vec{B}| \cos \theta \\ \Rightarrow 4 &= 3 \cdot 7 \cos \theta \\ \Rightarrow \theta &= \cos^{-1} \frac{4}{21} \\ \Rightarrow \theta &= 79^\circ\end{aligned}$$

Q2: Determine the value of m where

$\vec{A} = 2\hat{i} + m\hat{j} + \hat{k}$ and $\vec{B} = 4\hat{i} - 2\hat{j} - 2\hat{k}$ are perpendicular.

Solution: Since $\vec{A}$ and $\vec{B}$ are perpendicular, so

$$\begin{aligned}\vec{A} \cdot \vec{B} &= 0 \\ \Rightarrow (2\hat{i} + m\hat{j} + \hat{k}) \cdot (4\hat{i} - 2\hat{j} - 2\hat{k}) &= 0 \\ \Rightarrow 8 - 2m - 2 &= 0\end{aligned}$$

$$\Rightarrow 6 - 2m = 0$$

$$\Rightarrow 2m = 6$$

$$\Rightarrow m = 3 \text{ Ans.}$$

Q3: If $\bar{A} = 3\bar{i} - \bar{j} + 2\bar{k}$, $\bar{B} = 2\bar{i} + \bar{j} - \bar{k}$ and $\bar{C} = \bar{i} - 2\bar{j} + 2\bar{k}$ then show that $(\bar{A} \times \bar{B}) \times \bar{C} \neq \bar{A} \times (\bar{B} \times \bar{C})$.

Solution: $\bar{A} \times \bar{B} = \begin{vmatrix} \bar{i} & \bar{j} & \bar{k} \\ 3 & -1 & 2 \\ 2 & 1 & -1 \end{vmatrix}$

$$= -\bar{i} + 7\bar{j} + 5\bar{k}$$

$$\therefore (\bar{A} \times \bar{B}) \times \bar{C} = \begin{vmatrix} \bar{i} & \bar{j} & \bar{k} \\ -1 & 7 & 5 \\ 1 & -2 & 2 \end{vmatrix}$$

$$= 24\bar{i} + 7\bar{j} - 5\bar{k}$$

Again,

$$\bar{B} \times \bar{C} = \begin{vmatrix} \bar{i} & \bar{j} & \bar{k} \\ 2 & 1 & -1 \\ 1 & -2 & 2 \end{vmatrix}$$

$$= \bar{i} - 5\bar{j} - 5\bar{k}$$

$$\therefore \bar{A} \times (\bar{B} \times \bar{C}) = \begin{vmatrix} \bar{i} & \bar{j} & \bar{k} \\ 3 & -1 & 2 \\ 0 & -5 & -5 \end{vmatrix}$$

$$= 15\hat{i} - 15\hat{j} - 15\hat{k}$$

$$\therefore \text{L.H.S.} \neq \text{R.H.S.}$$

Q4: Find the unit vector perpendicular to each of the vectors $\vec{A} = 2\hat{i} + \hat{j} + \hat{k}$ and $\vec{B} = \hat{i} - \hat{j} + 2\hat{k}$.

Solution: Here,

$$\vec{A} \times \vec{B} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 1 & 1 \\ 1 & -1 & 2 \end{vmatrix}$$

$$= 3\hat{i} - 3\hat{j} - 3\hat{k}$$

$$\begin{aligned} \therefore |\vec{A} \times \vec{B}| &= \sqrt{3^2 + (-3)^2 + (-3)^2} \\ &= \sqrt{27} \\ &= 3\sqrt{3} \end{aligned}$$

$$\begin{aligned} \therefore \text{the unit vector, } \underline{e} &= \frac{\vec{A} \times \vec{B}}{|\vec{A} \times \vec{B}|} \\ &= \frac{3(\hat{i} - \hat{j} - \hat{k})}{3\sqrt{3}} \\ &= \frac{\hat{i} - \hat{j} - \hat{k}}{\sqrt{3}} \quad \text{Ans.} \end{aligned}$$

Q5: Find the projection of the vector $\vec{A} = 2\hat{i} - 3\hat{j} - \hat{k}$ on the vector $\vec{B} = \hat{i} + 4\hat{j} - 3\hat{k}$.

(2)

Solution: Hence,

$$\begin{aligned}\text{Unit vector, } \underline{e} &= \frac{\underline{b}}{|\underline{b}|} \\ &= \frac{\underline{i} + 4\underline{j} - 2\underline{k}}{\sqrt{1^2 + 4^2 + 2^2}} \\ &= \frac{\underline{i} + 4\underline{j} - 2\underline{k}}{\sqrt{21}}\end{aligned}$$

$\therefore$ Projection of $\underline{A}$ on $\underline{B} = \underline{A} \cdot \underline{e}$

$$= (2\underline{i} - 3\underline{j} - \underline{k}) \cdot \left(\frac{\underline{i} + 4\underline{j} - 2\underline{k}}{\sqrt{21}} \right)$$

$$= \frac{2 - 12 + 2}{\sqrt{21}}$$

$$= -\frac{8}{\sqrt{21}} \text{ Ans.}$$

Q. Vector differentiation:

Q1: If $\underline{r} = \sin t \underline{i} + \cos t \underline{j} + t \underline{k}$ then find

$$\text{(i) } \frac{d\underline{r}}{dt} \quad \text{(ii) } \frac{d^2\underline{r}}{dt^2} \quad \text{(iii) } \left| \frac{d\underline{r}}{dt} \right| \quad \text{(iv) } \left| \frac{d^2\underline{r}}{dt^2} \right|$$

Solution: Given,

$$\underline{r} = \sin t \underline{i} + \cos t \underline{j} + t \underline{k}$$

$$\text{(i) } \frac{d\underline{r}}{dt} = \cos t \underline{i} - \sin t \underline{j} + \underline{k}$$

$$\text{(ii) } \frac{d^2\underline{r}}{dt^2} = -\sin t \underline{i} - \cos t \underline{j}$$

$$\begin{aligned}\text{(iii) } \left| \frac{d\underline{r}}{dt} \right| &= \sqrt{\cos^2 t + (-\sin t)^2 + 1} \\ &= \sqrt{2}\end{aligned}$$

$$\textcircled{\text{iv}} \left| \frac{d^2 \vec{r}}{dt^2} \right| = \sqrt{\sin^2 t + \cos^2 t} = 1$$

$$\text{and } \vec{B} = \sin t \underline{i} - \cos t \underline{j}$$

Q2: IF $\vec{A} = 5t^2 \underline{i} + t \underline{j} - t^3 \underline{k}$ then find

$$\textcircled{\text{i}} \frac{d}{dt} (\vec{A} \cdot \vec{B})$$

$$\textcircled{\text{ii}} \frac{d}{dt} (\vec{A} \times \vec{B})$$



Solution: Here,

$$\vec{A} \cdot \vec{B} = 5t^2 \sin t - t \cos t$$

$$\therefore \frac{d}{dt} (\vec{A} \cdot \vec{B}) = 5t^2 \cdot \cos t + \sin t \cdot 10t + t \sin t - \cos t$$

$$= 5t^2 \cos t + 10t \sin t + t \sin t - \cos t$$

$$\textcircled{\text{ii}} \vec{A} \times \vec{B} = \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ 5t^2 & t & -t^3 \\ \sin t & -\cos t & 0 \end{vmatrix}$$

$$= -\cos t \cdot t^3 \underline{i} - \sin t \cdot t^3 \underline{j} + \underline{k} (-5t^2 \cos t - t \sin t)$$

$$\therefore \frac{d}{dt} (\vec{A} \times \vec{B}) = (\sin t \cdot t^3 - 3t^2 \cos t) \underline{i} - (3t^2 \sin t + t^3 \cos t) \underline{j} + (-10t \cos t + 5t^2 \sin t - \sin t + t \cos t) \underline{k}$$

$$= (t^3 \sin t - 3t^2 \cos t) \underline{i} - (3t^2 \sin t + t^3 \cos t) \underline{j} + (5t^2 \sin t - \sin t - 9t \cos t) \underline{k}$$

Gradient, Divergence, Curl:

Del: The vector differential operator Del, written by ∇ , is defined by

$$\nabla = \frac{\partial}{\partial x} \underline{i} + \frac{\partial}{\partial y} \underline{j} + \frac{\partial}{\partial z} \underline{k}$$

Gradient: Let $\phi(x, y, z)$ be a differentiable scalar function at all point in a region then the gradient of ϕ , written $\text{grad } \phi$ or $\nabla \phi$ is defined by -

$$\begin{aligned}\text{grad } \phi = \nabla \phi &= \left(\frac{\partial}{\partial x} \underline{i} + \frac{\partial}{\partial y} \underline{j} + \frac{\partial}{\partial z} \underline{k} \right) \phi \\ &= \frac{\partial \phi}{\partial x} \underline{i} + \frac{\partial \phi}{\partial y} \underline{j} + \frac{\partial \phi}{\partial z} \underline{k}\end{aligned}$$

Divergence: Let $\underline{v}(x, y, z) = v_1 \underline{i} + v_2 \underline{j} + v_3 \underline{k}$ be defined and differentiable at each point (x, y, z) . Then the divergence of $\underline{v}$, written $\text{div } \underline{v}$ or $\nabla \cdot \underline{v}$ is defined by

$$\begin{aligned}\nabla \cdot \underline{v} &= \left(\frac{\partial}{\partial x} \underline{i} + \frac{\partial}{\partial y} \underline{j} + \frac{\partial}{\partial z} \underline{k} \right) \cdot (v_1 \underline{i} + v_2 \underline{j} + v_3 \underline{k}) \\ &= \frac{\partial v_1}{\partial x} + \frac{\partial v_2}{\partial y} + \frac{\partial v_3}{\partial z}\end{aligned}$$

Curl: Let $\underline{v}(x, y, z) = v_1 \underline{i} + v_2 \underline{j} + v_3 \underline{k}$ be defined and differentiable at each point (x, y, z) . Then the curl of $\underline{v}$, written $\text{curl } \underline{v}$

or $\nabla \times \underline{v}$ is defined by

$$\nabla \times \underline{v} = \left(\frac{\partial}{\partial x} \underline{i} + \frac{\partial}{\partial y} \underline{j} + \frac{\partial}{\partial z} \underline{k} \right) \times (v_1 \underline{i} + v_2 \underline{j} + v_3 \underline{k})$$

$$= \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ v_1 & v_2 & v_3 \end{vmatrix}$$

Q1: If $\phi = 3xyz - x^2y^2z^2$ then find $\nabla \phi$ at the point $(1, 1, 1)$.

Solution: Here,

$$\phi = 3xyz - x^2y^2z^2$$

$$\begin{aligned} \therefore \nabla \phi &= \underline{i} \frac{\partial}{\partial x} (3xyz - x^2y^2z^2) + \underline{j} \frac{\partial}{\partial y} (3xyz - x^2y^2z^2) + \underline{k} \frac{\partial}{\partial z} (3xyz - x^2y^2z^2) \\ &= \underline{i} (3yz - 2xy^2z^2) + \underline{j} (3xz - 2x^2yz^2) + \underline{k} (3xy - 2x^2y^2z) \end{aligned}$$

$\therefore$ At the point $(1, 1, 1)$

$$\begin{aligned} \nabla \phi &= \underline{i} (3 - 2) + \underline{j} (3 - 2) + \underline{k} (3 - 2) \\ &= \underline{i} + \underline{j} + \underline{k} \quad \text{Ans.} \end{aligned}$$

Q2: If $\bar{A} = x^2z \underline{i} - 2y^3z^2 \underline{j} + xy^2z \underline{k}$ then find $\text{div } \bar{A}$ at the point $(1, -1, 1)$.

Solution: $\nabla \cdot \bar{A} = \frac{\partial}{\partial x} (x^2z) + \frac{\partial}{\partial y} (-2y^3z^2) + \frac{\partial}{\partial z} (xy^2z)$

$$= 2xz - 6y^2z^2 + xy^2$$

At the point $(1, -1, 1)$

$$\nabla \cdot \vec{A} = 2 \cdot 1 \cdot 1 - 6 \cdot 1^2 \cdot (-1)^2 + 1 \cdot (-1)^2 \\ = -3 \text{ Ans.}$$

Q3: If $\vec{A} = xz^3 \underline{i} - 2x^2yz \underline{j} + 2yz^4 \underline{k}$ then find $\text{curl } \vec{A}$ at the point $(1, -1, 1)$.

Solution:

$$\nabla \times \vec{A} = \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ xz^3 & -2x^2yz & 2yz^4 \end{vmatrix}$$

$$= \underline{i} \left[\frac{\partial}{\partial y} (2yz^4) + \frac{\partial}{\partial z} (2x^2yz) \right] \\ - \underline{j} \left[\frac{\partial}{\partial x} (2yz^4) - \frac{\partial}{\partial z} (xz^3) \right] + \underline{k} \left[\frac{\partial}{\partial x} (-2x^2yz) - \frac{\partial}{\partial y} (xz^3) \right] \\ = \underline{i} (2z^4 + 2x^2z) + \underline{j} \cdot 3xz^2 - \underline{k} \cdot 4xyz$$

∴ At the point $(1, -1, 1)$

$$\nabla \times \vec{A} = 3\underline{j} + 4\underline{k} \text{ Ans.}$$

□ A vector $\underline{v}$ is called solenoidal if $\nabla \cdot \underline{v} = 0$ / $\text{Div } \underline{v} = 0$

□ A vector $\underline{v}$ is called irrotational if $\text{Curl } \underline{v} = 0$ / $\nabla \times \underline{v} = 0$

□ A vector $\underline{v}$ is called rotational if $\text{Curl } \underline{v} \neq 0$ / $\nabla \times \underline{v} \neq 0$.

Q1: Show that $\underline{f} = (x+3y)\underline{i} + (y-3z)\underline{j} + (x-2z)\underline{k}$ is solenoidal.

Solution: Here, $\underline{f} = (x+3y)\underline{i} + (y-3z)\underline{j} + (x-2z)\underline{k}$

$$\therefore \nabla \cdot \underline{f} = \frac{\partial}{\partial x}(x+3y) + \frac{\partial}{\partial y}(y-3z) + \frac{\partial}{\partial z}(x-2z)$$
$$= 1 + 1 - 2$$
$$= 0$$

$\therefore$ Since $\nabla \cdot \underline{f} = 0$, so $\underline{f}$ is solenoidal.

Q2: If $\underline{F}(x, y, z) = (x+3y)\underline{i} + (y-2z)\underline{j} + (x+az)\underline{k}$ is solenoidal then find the value of a .

Solution: Since $\underline{F}$ is solenoidal hence

$$\text{div } \underline{F} = 0$$
$$\Rightarrow \frac{\partial}{\partial x}(x+3y) + \frac{\partial}{\partial y}(y-2z) + \frac{\partial}{\partial z}(x+az) = 0$$

$$\Rightarrow 1 + 1 + a = 0$$

$$\Rightarrow a = -2$$

Ans.

Q3: show that $\underline{f} = (\sin y + z)\underline{i} + (x \cos y - z)\underline{j} + (x - y)\underline{k}$ is irrotational.

Solution: Here

$$\begin{aligned}\nabla \times \underline{f} &= \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \sin y + z & x \cos y - z & x - y \end{vmatrix} \\ &= \underline{i} \left[\frac{\partial}{\partial y} (x - y) - \frac{\partial}{\partial z} (x \cos y - z) \right] \\ &\quad - \underline{j} \left[\frac{\partial}{\partial x} (x - y) - \frac{\partial}{\partial z} (\sin y + z) \right] + \underline{k} \left[\frac{\partial}{\partial x} (x \cos y - z) - \frac{\partial}{\partial y} (\sin y + z) \right] \\ &= (-1 + 1)\underline{i} - \underline{j} (1 - 1) + \underline{k} (\cos y - \cos y) \\ &= 0\end{aligned}$$

Since $\text{curl } \underline{f} = 0$, so $\underline{f}$ is irrotational.

Q4: If $\underline{F}(x, y, z) = (x + 2y + az)\underline{i} + (bx - 3y - z)\underline{j} + (4z + cy + 2x)\underline{k}$ is irrotational then find the value of a, b, c .

Solution: Since $\underline{F}$ is irrotational, so

$$\begin{aligned}\nabla \times \underline{F} &= 0 \\ \Rightarrow \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x + 2y + az & bx - 3y - z & 4z + cy + 2x \end{vmatrix} &= 0\end{aligned}$$

$$\Rightarrow \left[\frac{\partial}{\partial y}(4x + cy + 2z) - \frac{\partial}{\partial z}(bx - 3y - z) \right] \underline{i} - \underline{j} + \left[\frac{\partial}{\partial y}(4x + cy + 2z) - \frac{\partial}{\partial z}(x + 2y + az) \right] + \left[\frac{\partial}{\partial x}(bx - 3y - z) - \frac{\partial}{\partial y}(x + 2y + az) \right] \underline{k} = 0$$

$$\Rightarrow (c+1)\underline{i} + (a-4)\underline{j} + (b-2)\underline{k} = 0$$

$$\Rightarrow c+1=0, \quad a-4=0, \quad b-2=0$$

$$\Rightarrow c=-1, \quad a=4, \quad b=2$$

Ans.

Q5: IF $\phi = 2x^3y^2z^4$ then find $\text{div grad } \phi / (\nabla \cdot \nabla \phi)$.

Solution:

$$\begin{aligned} \text{grad } \phi &= \frac{\partial}{\partial x}(2x^3y^2z^4)\underline{i} + \frac{\partial}{\partial y}(2x^3y^2z^4)\underline{j} + \frac{\partial}{\partial z}(2x^3y^2z^4)\underline{k} \\ &= 6x^2y^2z^4\underline{i} + 4x^3yz^4\underline{j} + 8x^3y^2z^3\underline{k} \end{aligned}$$

$$\begin{aligned} \therefore \text{div grad } \phi &= \left(\frac{\partial}{\partial x}\underline{i} + \frac{\partial}{\partial y}\underline{j} + \frac{\partial}{\partial z}\underline{k} \right) \cdot (6x^2y^2z^4\underline{i} + 4x^3yz^4\underline{j} + 8x^3y^2z^3\underline{k}) \\ &= \frac{\partial}{\partial x}(6x^2y^2z^4) + \frac{\partial}{\partial y}(4x^3yz^4) + \frac{\partial}{\partial z}(8x^3y^2z^3) \end{aligned}$$

$$= 12x^2y^2z^4 + 4x^3z^4 + 24x^3y^2z^2 \quad \text{Ans.}$$

Q6: If $\vec{A} = x^2y\hat{i} - 2xz\hat{j} + 2yz\hat{k}$ then find $\text{Curl } \vec{A} / \nabla \times (\nabla \times \vec{A})$

Solution: Here,

$$\text{Curl } \vec{A} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^2y & -2xz & 2yz \end{vmatrix}$$

$$= \left[\frac{\partial}{\partial y} (2yz) + \frac{\partial}{\partial z} (2xz) \right] \hat{i} - \left[\frac{\partial}{\partial z} (2yz) - \frac{\partial}{\partial z} (x^2y) \right] \hat{j} + \left[\frac{\partial}{\partial x} (-2xz) - \frac{\partial}{\partial y} (x^2y) \right] \hat{k}$$

$$= (2x + 2z) \hat{i} + (-x^2 - 2z) \hat{k}$$

$$\therefore \text{Curl } \text{Curl } \vec{A} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2x+2z & 0 & (-x^2-2z) \end{vmatrix}$$

$$= \left[\frac{\partial}{\partial y} (-x^2 - 2z) \right] \hat{i} - \hat{j} \left[\frac{\partial}{\partial x} (-x^2 - 2z) \right] - \frac{\partial}{\partial z} (2x + 2z) \hat{k}$$

$$= (2 + 2x) \hat{j} \quad \text{Ans.}$$

H.W: If $\vec{F} = 5x^2\hat{i} + 2y^2\hat{j} + 2z^2\hat{k}$ then find $\nabla \cdot (\nabla \times \vec{F})$.

Vector Integration

Q IF $\underline{r} = t\hat{i} - t^2\hat{j} + (t-1)\hat{k}$ and $\underline{s} = 2t^2\hat{i} + 6t\hat{k}$
then evaluate -

(i) $\int_0^2 \underline{r} \cdot \underline{s} \, dt$ (ii) $\int_0^2 \underline{r} \times \underline{s} \, dt$

Solution: (i) Here

$$\begin{aligned}\underline{r} \cdot \underline{s} &= (t\hat{i} - t^2\hat{j} + (t-1)\hat{k}) \cdot (2t^2\hat{i} + 6t\hat{k}) \\ &= 2t^3 + 6t^2 - 6t\end{aligned}$$

$$\therefore \int_0^2 \underline{r} \cdot \underline{s} \, dt = \int_0^2 (2t^3 + 6t^2 - 6t) \, dt$$

$$= \left[\frac{t^4}{2} + 2t^3 - 3t^2 \right]_0^2$$

$$= 8 + 16 - 12$$

$$= 12 \text{ Ans.}$$

(ii) Here

$$\underline{r} \times \underline{s} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ t & -t^2 & t-1 \\ 2t^2 & 0 & 6t \end{vmatrix}$$

$$= -6t^3\hat{i} - (8t^2 - 2t^3)\hat{j} + 2t^4\hat{k}$$

$$\therefore \int_0^2 \underline{r} \times \underline{s} \, dt = \int_0^2 \left[-6t^3\hat{i} - (8t^2 - 2t^3)\hat{j} + 2t^4\hat{k} \right] dt$$

$$= -6\hat{i} \int_0^2 t^3 \, dt - \hat{j} \int_0^2 (8t^2 - 2t^3) \, dt$$

$$+ 2k \int_0^2 t^4 dt$$

$$= -6i \left[\frac{t^4}{4} \right]_0^2 - j \left[\frac{8t^3}{3} - \frac{2t^4}{4} \right]_0^2 + 2k \left[\frac{t^5}{5} \right]_0^2$$

$$= -24i - \frac{40}{3}j + \frac{64}{5}k \text{ Ans.}$$

Line Integral: Let $\underline{F}(\underline{r})$ be a continuous vector point function defined on smooth curve C given by $\underline{r} = \underline{f}(t)$. If s denotes the arc length of any point $P(x, y, z)$ on the curve from a fixed point then

$$\int_C \underline{F} \cdot \frac{d\underline{r}}{ds} ds \text{ or } \int_C \underline{F} \cdot d\underline{r}$$

is called line integral of $\underline{F}(\underline{r})$ along C .

Surface integral: Let $\underline{F}(\underline{r})$ be a continuous vector point function and $\underline{r} = \underline{f}(u, v)$ be a smooth ~~curve~~ surface S . If $\underline{n}$ is a unit vector then $\iint_S \underline{F} \cdot \underline{n} ds$ is called surface integral.

Volume Integral: If V is the volume of a surface then $\iiint_V \underline{A} dv$ is called volume integral where $\underline{A}$ is a vector function.

Q1: IF $\underline{F} = 3xy^2 \underline{i} + y^2 \underline{j}$ then evaluate $\int_c \underline{F} \cdot d\underline{r}$ where c is the curve $y = 2x^2$ in the xy -plane from $(0, 0)$ to $(1, 2)$.

Solution: Given,

$$\underline{F} = 3xy^2 \underline{i} + y^2 \underline{j}$$

$$d\underline{r} = dx \underline{i} + dy \underline{j} + dz \underline{k}$$

Also $y = 2x^2$

$$\Rightarrow dy = 4x dx$$

$$\therefore \int_c \underline{F} \cdot d\underline{r} = \int_c (3xy^2 \underline{i} + y^2 \underline{j}) \cdot (dx \underline{i} + dy \underline{j} + dz \underline{k})$$

$$= \int_c (3xy^2 dx + y^2 dy)$$

$$= \int_0^1 (3x \cdot 4x^4 dx + 4x^2 \cdot 4x dx)$$

$$= \int_0^1 (12x^5 dx + 16x^3 dx)$$

$$= \left[12 \cdot \frac{x^6}{6} + 16 \cdot \frac{x^4}{4} \right]_0^1$$

$$= 2 + 4$$

$$= 6 \text{ Ans.}$$

Q2: IF $\underline{F} = xy\underline{i} - z\underline{j} + x^2\underline{k}$ then find

(i) $\int_c \underline{F} \cdot d\underline{r}$ (ii) $\int_c \underline{F} \times d\underline{r}$ where c is the curve $x=t^2$, $y=2t$, $z=t^3$ from $t=0$ to $t=1$.

Solution: Here,

$$\begin{aligned}\underline{F} &= xy\underline{i} - z\underline{j} + x^2\underline{k} \\ &= 2t^3\underline{i} - t^3\underline{j} + t^4\underline{k}\end{aligned}$$

$$\begin{aligned}d\underline{r} &= x\underline{i} + y\underline{j} + z\underline{k} \\ &= t^2\underline{i} + 2t\underline{j} + t^3\underline{k}\end{aligned}$$

$$\begin{aligned}\therefore d\underline{r} &= 2t dt \underline{i} + 2 dt \underline{j} + 3t^2 dt \underline{k} \\ &= (2t\underline{i} + 2\underline{j} + 3t^2\underline{k}) dt\end{aligned}$$

$$\begin{aligned}\therefore \int_c \underline{F} \cdot d\underline{r} &= \int_0^1 (4t^4 - 2t^3 + 3t^6) dt \\ &= \left[4 \frac{t^5}{5} - 2 \frac{t^4}{4} + 3 \frac{t^7}{7} \right]_0^1 \\ &= \frac{4}{5} - \frac{1}{2} + \frac{3}{7} \\ &= \frac{51}{70}\end{aligned}$$

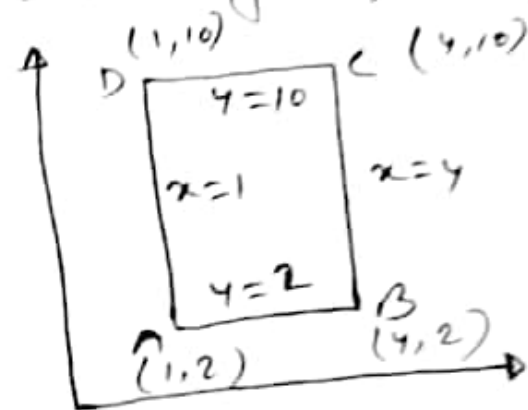
$$\begin{aligned}\text{(ii)} \quad \underline{F} \times d\underline{r} &= \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ 2t^3 & -t^3 & t^4 \\ 2t & 2 & 3t^2 \end{vmatrix} dt \\ &= + \left[-(3t^5 + 2t^4) \underline{i} - 4t^5 \underline{j} + (4t^{\frac{3}{2}} + 2t^4) \underline{k} \right] dt\end{aligned}$$

$$\therefore \int_C \underline{F} \times d\underline{r} = \int_0^1 [-(3+t^5+2+t^4)\underline{i} - 4+t^5\underline{j} + (4+t^3+2+t^4)\underline{k}] dt$$

$$= -\left(\frac{1}{2} + \frac{2}{5}\right)\underline{i} - \frac{2}{3}\underline{j} + \left(1 + \frac{2}{5}\right)\underline{k}$$

$$= -\frac{9}{10}\underline{i} - \frac{2}{3}\underline{j} + \frac{7}{5}\underline{k} \text{ Ans.}$$

Q3: Evaluate $\int_C \underline{F} \cdot d\underline{r}$ where $\underline{F} = xy\underline{i} + (x^2+y^2)\underline{j}$ and C is the rectangle in the xy plane bounded by the lines $y=2$, $x=4$, $y=10$ and $x=1$.



Solution: Here

$$\underline{F} = xy\underline{i} + (x^2+y^2)\underline{j}$$

$$\underline{r} = x\underline{i} + y\underline{j} + z\underline{k}$$

$$\therefore \underline{F} \cdot d\underline{r} = xy dx + (x^2+y^2) dy$$

Let C be the rectangle $ABCD$ shown in the figure.

$$\therefore \int_C \underline{F} \cdot d\underline{r} = \int_{AB} \underline{F} \cdot d\underline{r} + \int_{BC} \underline{F} \cdot d\underline{r} + \int_{CD} \underline{F} \cdot d\underline{r} + \int_{DA} \underline{F} \cdot d\underline{r}$$

Now, on AB , $y=2$, $dy=0$

$$\therefore \int_{AB} \underline{F} \cdot d\underline{r} = \int_{AB} xy dx + (x^2+y^2) dy$$

$$= \int_1^4 2x \, dx$$

$$= 2 \left[\frac{x^2}{2} \right]_1^4$$

$$= 4^2 - 1$$

$$= 15$$

On BC, $x=4$, $dx=0$

$$\therefore \int_{BC} \underline{F} \cdot d\underline{r} = \int_2^{10} (16 + y^2) \, dy$$

$$= \left[16y + \frac{y^3}{3} \right]_2^{10}$$

$$= 160 + \frac{1000}{3} - 32 - \frac{8}{3}$$

$$= 128 + \frac{992}{3}$$

On CD, $y=10$ so that $dy=0$

$$\therefore \int_{CD} \underline{F} \cdot d\underline{r} = \int_{CD} xy \, dx + (x^2 + y^2) \, dy$$

$$= \int_4^1 10x \, dx$$

$$= 5 \left[x^2 \right]_4^1$$

$$= -75$$

and on DA, $x=1$, $dx=0$

$$\therefore \int_{DA} \underline{F} \cdot d\underline{r} = \int_{10}^2 (1+y^2) dy$$

$$= \left[y + \frac{y^3}{3} \right]_{10}^2$$

$$= 2 + \frac{8}{3} - 10 - \frac{1000}{3}$$

$$= -8 - \frac{992}{3}$$

$$\therefore \int_C \underline{F} \cdot d\underline{r} = 15 + 128 + \frac{992}{3} - 75 - 8 - \frac{992}{3}$$

$$= 60 \text{ Ans.}$$

Q4: ~~evaluate~~ show that $\underline{F} = (2xy + z^3)\underline{i} + x^2\underline{j} + 3xz^2\underline{k}$ is a conservative force field and hence find the work done in moving an object in this field $(1, -2, 1)$ to $(3, 1, 4)$.

Solution: Given,

$$\underline{F} = (2xy + z^3)\underline{i} + x^2\underline{j} + 3xz^2\underline{k}$$

$$\therefore \nabla \times \underline{F} = \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2xy + z^3 & x^2 & 3xz^2 \end{vmatrix}$$

$$= \underline{i}(0 - 0) - \underline{j}(3z^2 - 3z^2) + \underline{k}(2x - 2x)$$

$$= 0$$

$\therefore \underline{F}$ is a conservative force field.

Now the work done in moving an o_2 in this field is

$$W = \int_{(1, -2, 1)}^{(3, 1, 4)} \underline{f} \cdot d\underline{r}$$

Now, $\underline{f} \cdot d\underline{r} = (2xy + z^3)dx + x^2dy + 3xz^2dz$
 $\therefore$ The straight line through $(1, -2, 1)$ and $(3, 1, 4)$ is

$$\frac{x-1}{1-3} = \frac{y+2}{-2-1} = \frac{z-1}{1-4}$$

$$\Rightarrow \frac{x-1}{2} = \frac{y+2}{3} = \frac{z-1}{3} = t \text{ (say)}$$

$$\Rightarrow x = 2t+1, \quad y = 3t-2, \quad z = 3t+1 \quad \text{where} \\ 0 \leq t \leq 1.$$

$$\begin{aligned} \therefore \underline{f} \cdot d\underline{r} &= [2(2t+1)(3t-2) + (3t+1)^3] d(2t+1) \\ &\quad + (2t+1)^2 d(3t-2) + 3(2t+1)(3t+1)^2 d(3t+1) \\ &= (12t^2 - 2t - 4 + 27t^3 + 27t^2 + 9t + 1) \cdot 2dt \\ &\quad + (4t^2 + 4t + 1) \cdot 3dt + 3(18t^3 + 21t^2 + 8t + 1) \cdot 3dt \\ &= (24t^3 + 279t^2 + 98t + 6) dt \end{aligned}$$

$$\begin{aligned} \therefore W &= \int_0^1 (24t^3 + 279t^2 + 98t + 6) dt \\ &= [54t^4 + 93t^3 + 49t^2 + 6t]_0^1 \end{aligned}$$

$$= 54 + 93 + 49 + 6$$

$$= 202$$

Ans.

Q5: Evaluate $\iint_S \underline{F} \cdot \underline{n} \, ds$ where $\underline{F} = x^2 \underline{i} + y^2 \underline{j} + z^2 \underline{k}$ and S is that portion of the plane $x+y+z=1$ which lies in the first octant.

Solution: Here

$$\iint_S \underline{F} \cdot \underline{n} \, ds = \iint_R \underline{F} \cdot \underline{n} \frac{dx \, dy}{|\underline{n} \cdot \underline{k}|}$$

$$\text{Let, } \phi(x, y, z) = x + y + z - 1$$

$$\therefore \underline{\nabla} \phi = \underline{i} + \underline{j} + \underline{k}$$

$$\therefore |\underline{\nabla} \phi| = \sqrt{3}$$

$$\therefore \underline{n} = \frac{\underline{\nabla} \phi}{|\underline{\nabla} \phi|}$$

$$= \frac{1}{\sqrt{3}} (\underline{i} + \underline{j} + \underline{k})$$

$$\begin{aligned} \therefore \underline{n} \cdot \underline{k} &= \frac{1}{\sqrt{3}} (\underline{i} + \underline{j} + \underline{k}) \cdot \underline{k} \\ &= \frac{1}{\sqrt{3}} \end{aligned}$$

$$\therefore \underline{F} \cdot \underline{n} = \frac{1}{\sqrt{3}} (x^2 + y^2 + z^2)$$

$$= \frac{1}{\sqrt{3}} (x^2 + y^2 + (1-x-y)^2) \quad [\because z = 1-x-y]$$

$$\begin{aligned}
 \therefore \iint_S \underline{F} \cdot \underline{n} \, ds &= \iint_R \frac{1}{\sqrt{3}} (x^2 + y^2 + (1-x^2-y^2)^2) \frac{1}{\sqrt{3}} \, dx \, dy \\
 &= \int_{x=0}^1 \int_{y=0}^{1-x} \left[x^2 + y^2 + (1-x^2-y^2)^2 \right] dx \, dy \\
 &= \int_0^1 \left[x^2 y + \frac{y^3}{3} + \frac{(1-x^2-y^2)^3}{3(-1)} \right]_0^{1-x} dx \\
 &= \int_0^1 \left[x^2(1-x) + \frac{1}{3}(1-x)^3 - \frac{1}{3}(1-x)^3 \right] dx \\
 &= \left[\frac{x^3}{3} - \frac{x^4}{4} + \frac{1}{3} \frac{(1-x)^4}{4(-1)} - \frac{1}{3} \frac{(1-x)^4}{4(-1)} \right]_0^1 \\
 &= \frac{1}{6} \text{ Ans. } \frac{1}{6}
 \end{aligned}$$

Q6: Show that $\iint_S \underline{A} \cdot \underline{n} \, ds = 81$ where $\underline{A} = (x+y^2)\underline{i} + (-xz)\underline{j} + 2yz\underline{k}$ where, S is the surface of the plane $2x+y+2z=6$ in the first octant.

Solution: Here,

$$\iint_S \underline{A} \cdot \underline{n} \, ds = \iint_R \underline{A} \cdot \underline{n} \frac{dx \, dy}{|\underline{n} \cdot \underline{k}|}$$

Let $\phi(x, y, z) = 2x + y + 2z - 6$

$$\therefore \underline{n} = \frac{\nabla \phi}{|\nabla \phi|} = \frac{2\underline{i} + \underline{j} + 2\underline{k}}{3} \quad \therefore \underline{n} \cdot \underline{k} = \frac{2}{3}$$

$$\begin{aligned}\underline{A} \cdot \underline{n} &= \frac{2(x+y^2) - 2x + 4y^2}{3} \\ &= \frac{2y^2 + 2y(6-2x-y)}{3} \\ &= \frac{4(3-x)y}{3}\end{aligned}$$

$$\begin{aligned}\therefore \iint_S \underline{A} \cdot \underline{n} \, ds &= \iint_R \frac{4(3-x)y}{3} \cdot \frac{dx \, dy}{\frac{2}{3}} \\ &= 2 \int_0^3 \int_0^{6-2x} (3-x)y \, dx \, dy \\ &= 2 \int_0^3 (3-x) \left[\frac{y^2}{2} \right]_0^{6-2x} dx \\ &= \int_0^3 (3-x)(6-2x)^2 dx \\ &= 4 \int_0^3 (3-x)^3 dx \\ &= 4 \left[-\frac{(3-x)^4}{4} \right]_0^3 \\ &= 81 \text{ Ans.}\end{aligned}$$

Q7: Find the value of $\iiint_V \underline{F} \, dV$ where $\underline{F} = 2xz\mathbf{i} - x\mathbf{j} + y^2\mathbf{k}$ and V is the region bounded by the ~~curved~~ surfaces $x=0, y=0$

$$y=6, z=x^2, z=4.$$

Solution: when $z=4$ then $x=2$

$$\begin{aligned} \therefore \iiint_V \underline{F} dV &= \iiint_V (2xz \underline{i} - x \underline{j} + y^2 \underline{k}) dx dy dz \\ &= \int_0^2 \int_{y=0}^6 \int_{z=x^2}^4 (2xz \underline{i} - x \underline{j} + y^2 \underline{k}) dz dy dx \\ &= \int_0^2 \int_0^6 \left[xz^2 \underline{i} - xz \underline{j} + y^2 z \underline{k} \right]_{x^2}^4 dy dx \\ &= \int_0^2 \int_0^6 \left[(16x - x^5) \underline{i} - (4x - x^3) \underline{j} + (4y^2 - x^2 y^2) \underline{k} \right] dy dx \\ &= \int_0^2 \left\{ (16x - x^5) \left[y \right]_0^6 \underline{i} - (4x - x^3) \left[y \right]_0^6 \underline{j} + \left[\frac{4}{3} y^3 - \frac{x^2 y^3}{3} \right]_0^6 \underline{k} \right\} dx \\ &= \int_0^2 \left[6(16x - x^5) \underline{i} - 6(4x - x^3) \underline{j} + (288 - 72x^2) \underline{k} \right] dx \\ &= 6 \underline{i} \left[8x^2 - \frac{x^6}{6} \right]_0^2 - 6 \underline{j} \left[2x^2 - \frac{x^3}{3} \right]_0^2 + \underline{k} \left[288x - \frac{72x^3}{3} \right]_0^2 \\ &= 128 \underline{i} - 24 \underline{j} + 384 \underline{k} \end{aligned}$$

Ans.

Q8: Find $\iiint_V \nabla \cdot \underline{f} \, dv$ where $\underline{f} = (2x^2 - 3z)\underline{i} - 2xy\underline{j} - 4xz\underline{k}$ and V is the region bounded by $x=0, y=0, z=0$ and $2x+2y+z=4$.

Solution:

$$\nabla \cdot \underline{f} = 4x - 2x = 2x$$

$$\therefore \iiint_V \nabla \cdot \underline{f} \, dv = \int_{x=0}^2 \int_{y=0}^{2-x} \int_{z=0}^{4-2x-2y} 2x \, dz \, dy \, dx$$

$$= 2 \int_0^2 \int_0^{2-x} x \left[z \right]_0^{4-2x-2y} \, dy \, dx$$

$$= 2 \int_0^2 \int_0^{2-x} (4x - 2x^2 - 2xy) \, dy \, dx$$

$$= 2 \int_0^2 \left[4xy - 2x^2y - xy^2 \right]_0^{2-x} \, dx$$

$$= 2 \int_0^2 (2x^3 - 8x^2 + 8x) \, dx$$

$$= \frac{8}{3} \text{ Ans.}$$

Green's Theorem: Let R is a closed region of the xy plane bounded by a simple closed curve C such that any straight line parallel to the co-ordinate axes cuts C at most two points. If $P(x, y)$ and $Q(x, y)$ are continuous function of x and y having continuous partial derivatives in R then

$$\oint_C P dx + Q dy = \iint_R \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx dy$$

Problem 1: Verify Green's theorem in the plane for $\oint (xy + y^2) dx + x^2 dy$ where C is the closed curve of the region bounded by $y = x$ and $y = x^2$.

Solution: For intersection points,

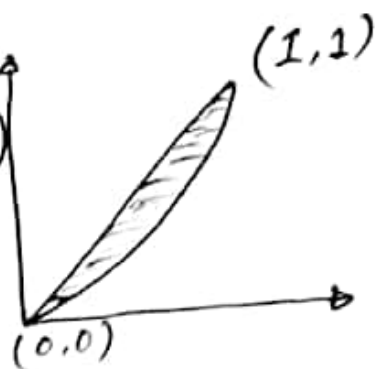
$$x = x^2$$

$$\Rightarrow x - x^2 = 0$$

$$\Rightarrow x(1 - x) = 0$$

$$\therefore x = 0, 1, \text{ and } y = 0, 1$$

Along $y = x^2$ the line integral from $(0,0)$ to $(1,1)$



$$\int_0^1 \left[(x \cdot x^2 + x^4) dx + x^2 dx^2 \right]$$

$$= \int_0^1 (x^3 + x^4 + 2x^3) dx$$

$$= \int_0^1 (3x^3 + x^4) dx$$

$$= \left[3 \frac{x^4}{4} + \frac{x^5}{5} \right]_0^1$$

$$= \frac{19}{20}$$

Again, along $y = x$ the line integral from $(1,1)$ to $(0,0)$ is,

$$\int_1^0 \left[(x \cdot x + x^2) dx + x^2 dx \right]$$

$$= \int_1^0 2x^2 dx$$

$$= -1$$

$$\therefore \text{the line integral is} = \frac{19}{20} - 1 = -\frac{1}{20}$$

Again, $P = xy + y^2$ and $Q = x^2$

$$\therefore \int \int_R \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx dy = \int \int_R (2x - x - 2y) dx dy$$

$$= \int_0^1 \int_{x^2}^x (x - 2y) dy dx$$

$$= \int_0^1 [xy - y^2]_{x^2}^x dx$$

$$= \int_0^1 (x^2 - x^2 - x^3 + x^4) dx$$

$$= \int_0^1 (x^4 - x^3) dx$$

$$= \left[\frac{x^5}{5} - \frac{x^4}{4} \right]_0^1$$

$$= -\frac{1}{20}$$

Hence, the green's theorem is verified.

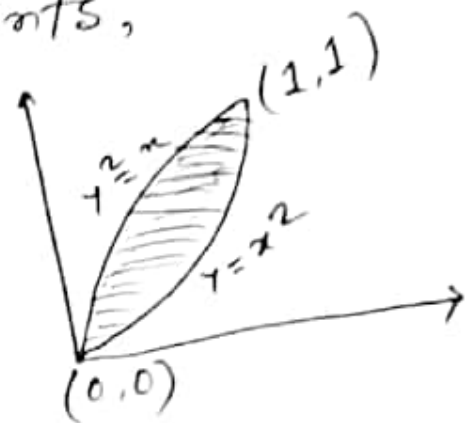
Problem 2: Verify Green's theorem for $\oint_C [(2xy - x^2)dx + (x + y^2)dy]$ where C is the closed curve of the region bounded by $y = x^2$ and $y^2 = x$.

Solution: For intersection points,

$$\begin{aligned}y^2 &= x \\ \Rightarrow (x^2)^2 &= x \\ \Rightarrow x^4 - x &= 0 \\ \Rightarrow x(x^3 - 1) &= 0\end{aligned}$$

$$\therefore x = 0, 1 \text{ and } y = 0, 1$$

$\therefore$ The intersection points are $(0, 0)$ and $(1, 1)$.



The line integral along $y = x^2$ from $(0, 0)$ to $(1, 1)$ is,

$$\begin{aligned}& \int_0^1 [(2x \cdot x^2 - x^2)dx + (x + x^4)2xdx] \\&= \int_0^1 (2x^3 + x^2 + 2x^5)dx \\&= \left[\frac{x^4}{2} + \frac{x^3}{3} + \frac{x^6}{3} \right]_0^1 \\&= \frac{7}{6}\end{aligned}$$

Again, the line integral along $y^2 = x$ from $(1,1)$ to $(0,0)$ is,

$$\int_1^0 \left[2 \cdot y^2 \cdot y - (y^2)^2 \cdot 2y dy + (y^2 + y^2) dy \right]$$

$$= \int_1^0 (4y^4 - 2y^5 + 2y^2) dy$$

$$= \left[\frac{4y^5}{5} - \frac{2y^6}{6} + \frac{2y^3}{3} \right]_1^0$$

$$= -\frac{17}{15}$$

$$\therefore \oint \left[(2xy - x^2) dx + (x + y^2) dy \right] = \frac{7}{6} - \frac{17}{15} = \frac{1}{30}$$

Again, $P = 2xy - x^2$ | $Q = x + y^2$
 $\Rightarrow \frac{\partial P}{\partial y} = 2x$ | $\Rightarrow \frac{\partial Q}{\partial x} = 1$

$$\therefore \iint_R \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx dy = \iint_R (1 - 2x) dx dy$$

$$= \int_0^1 \int_{x^2}^{\sqrt{x}} (1 - 2x) dx dy$$

$$= \int_0^1 (1 - 2x) [y]_{x^2}^{\sqrt{x}} dx$$

$$= \int_0^1 (1-2x) (\sqrt{x} - x^2) dx$$

$$= \int_0^1 (x^{\frac{1}{2}} - x^2 - 2x^{\frac{3}{2}} + 2x^3) dx$$

$$= \left[\frac{2}{3} x^{\frac{3}{2}} - \frac{x^3}{3} - \frac{4}{5} x^{\frac{5}{2}} + \frac{x^4}{2} \right]_0^1$$

$$= \frac{1}{30}.$$

Hence the Green's theorem is verified.

Supplementary
Vector Analysis

Gradient, Divergence, Curl

Find $\nabla \phi$ if (a) $\phi = \ln |\underline{r}|$, (b) $\phi = \frac{1}{r}$

Solution: (a) we know,

$$\underline{r} = x\underline{i} + y\underline{j} + z\underline{k}$$

$$\therefore |\underline{r}| = \sqrt{x^2 + y^2 + z^2}$$

$$\therefore \ln |\underline{r}| = \ln (x^2 + y^2 + z^2)^{\frac{1}{2}}$$
$$= \frac{1}{2} \ln (x^2 + y^2 + z^2)$$

$$\therefore \nabla \phi = \nabla (\ln |\underline{r}|)$$

$$= \nabla \left[\frac{1}{2} \ln (x^2 + y^2 + z^2) \right]$$

$$= \frac{1}{2} \left[\frac{\partial \ln (x^2 + y^2 + z^2)}{\partial x} \underline{i} + \frac{\partial \ln (x^2 + y^2 + z^2)}{\partial y} \underline{j} + \frac{\partial \ln (x^2 + y^2 + z^2)}{\partial z} \underline{k} \right]$$

$$= \frac{1}{2} \left[\frac{1}{x^2 + y^2 + z^2} \cdot 2x \underline{i} + \frac{1}{x^2 + y^2 + z^2} \cdot 2y \underline{j} + \frac{1}{x^2 + y^2 + z^2} \cdot 2z \underline{k} \right]$$

$$= \frac{1}{2} \left[\frac{2x \underline{i} + 2y \underline{j} + 2z \underline{k}}{x^2 + y^2 + z^2} \right]$$

$$= \frac{x \underline{i} + y \underline{j} + z \underline{k}}{x^2 + y^2 + z^2}$$

$$= \frac{\underline{r}}{r^2} \quad \text{Ans}$$

$$\textcircled{b} \nabla \phi = \nabla \left(\frac{1}{r} \right)$$

$$= \nabla \left(\frac{1}{\sqrt{x^2 + y^2 + z^2}} \right)$$

$$= \nabla \left\{ (x^2 + y^2 + z^2)^{-\frac{1}{2}} \right\}$$

$$= \underline{i} \frac{\partial}{\partial x} \left[(x^2 + y^2 + z^2)^{-\frac{1}{2}} \right] + \underline{j} \frac{\partial}{\partial y} \left[(x^2 + y^2 + z^2)^{-\frac{1}{2}} \right] + \underline{k} \frac{\partial}{\partial z} \left[(x^2 + y^2 + z^2)^{-\frac{1}{2}} \right]$$

$$= \underline{i} \left\{ -\frac{1}{2} (x^2 + y^2 + z^2)^{-\frac{3}{2}} \cdot 2x \right\} + \underline{j} \left\{ -\frac{1}{2} (x^2 + y^2 + z^2)^{-\frac{3}{2}} \cdot 2y \right\} + \underline{k} \left\{ -\frac{1}{2} (x^2 + y^2 + z^2)^{-\frac{3}{2}} \cdot 2z \right\}$$

$$= \frac{-x\underline{i} - y\underline{j} - z\underline{k}}{(x^2 + y^2 + z^2)^{3/2}}$$

$$= -\frac{\underline{r}}{r^3} \text{ Ans.}$$

Find a unit normal to the surface $x^2y + 2xz = 4$ at the point $(2, -2, 3)$.

Solution: Let,

$$\phi = x^2y + 2xz - 4$$

$$\therefore \nabla \phi = (2xy + 2z)\underline{i} + y\underline{j} + 2\underline{k}$$

At the point $(2, -2, 3)$

$$\nabla \phi = -2\mathbf{i} + 4\mathbf{j} + 4\mathbf{k}$$

$$\therefore |\nabla \phi| = \sqrt{4+16+16}$$

$$= 6$$

$\therefore$ Unit normal to the surface is

$$= \frac{\nabla \phi}{|\nabla \phi|}$$

$$= \frac{-2\mathbf{i} + 4\mathbf{j} + 4\mathbf{k}}{6}$$

$$= -\frac{1}{3}\mathbf{i} + \frac{2}{3}\mathbf{j} + \frac{2}{3}\mathbf{k} \text{ Ans.}$$

Find the directional derivative of $\phi = x^2yz + 4xz^2$ at $(1, -2, -1)$ in the direction $2\mathbf{i} - \mathbf{j} - 2\mathbf{k}$.

Solution: $\nabla \phi = (2xyz + 4z^2)\mathbf{i} + x^2z\mathbf{j} + \cancel{2x^2y} + (x^2y + 8xz)\mathbf{k}$

At the point $(1, -2, -1)$

$$\nabla \phi = 8\mathbf{i} - \mathbf{j} - 10\mathbf{k}$$

The unit vector in the direction of $2\mathbf{i} - \mathbf{j} - 2\mathbf{k}$ is $\underline{a} = \frac{2\mathbf{i} - \mathbf{j} - 2\mathbf{k}}{\sqrt{4+1+4}}$

$$= \frac{2}{3}\mathbf{i} - \frac{1}{3}\mathbf{j} - \frac{2}{3}\mathbf{k}$$

∴ the directional derivative is

$$= \nabla \phi \cdot a$$

$$= (8\hat{i} - \hat{j} - 10\hat{k}) \cdot \left(\frac{2}{3}\hat{i} - \frac{1}{3}\hat{j} - \frac{2}{3}\hat{k}\right)$$

$$= \frac{16}{3} + \frac{1}{3} + \frac{20}{3}$$

$$= \frac{37}{3} \text{ Ans.}$$

Q Find the angle between the surfaces $x^2 + y^2 + z^2 = 9$ and $z = x^2 + y^2 - 3$ at the point $(2, -1, 2)$.

Solution: Let,

$$\phi_1 = x^2 + y^2 + z^2 - 9$$

$$\therefore \nabla \phi_1 = 2x\hat{i} + 2y\hat{j} + 2z\hat{k}$$

$$\phi_2 = x^2 + y^2 - z - 3$$

$$\nabla \phi_2 = 2x\hat{i} + 2y\hat{j} - \hat{k}$$

→ At $(2, -1, 2)$, $\nabla \phi_1 = 4\hat{i} - 2\hat{j} + 4\hat{k}$, $\nabla \phi_2 = 4\hat{i} - 2\hat{j} - \hat{k}$

$$\therefore (\nabla \phi_1) \cdot (\nabla \phi_2) = |\nabla \phi_1| |\nabla \phi_2| \cos \theta$$

$$\Rightarrow \cos \theta = \frac{(4\hat{i} - 2\hat{j} + 4\hat{k}) \cdot (4\hat{i} - 2\hat{j} - \hat{k})}{\sqrt{16 + 4 + 16} \sqrt{16 + 4 + 1}}$$

$$\Rightarrow \cos \theta = \frac{16}{6\sqrt{21}}$$

$$\Rightarrow \theta = \cos^{-1} \frac{8}{3\sqrt{21}}$$

$$\Rightarrow \theta = 54.514^\circ \text{ Ans.}$$

Q Evaluate $\nabla \cdot (\underline{A} \times \underline{r})$ if $\nabla \times \underline{A} = 0$

Solution: Let,

$$\underline{A} = A_1 \underline{i} + A_2 \underline{j} + A_3 \underline{k}$$

$$\underline{r} = x \underline{i} + y \underline{j} + z \underline{k}$$

$$\therefore (\underline{A} \times \underline{r}) = \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ A_1 & A_2 & A_3 \\ x & y & z \end{vmatrix}$$

$$= \underline{i} (A_2 z - A_3 y) - \underline{j} (A_1 z - A_3 x) + \underline{k} (A_1 y - A_2 x)$$

$$\therefore \nabla \cdot (\underline{A} \times \underline{r}) = \frac{\partial}{\partial x} (A_2 z - A_3 y) - \frac{\partial}{\partial y} (A_1 z - A_3 x) + \frac{\partial}{\partial z} (A_1 y - A_2 x)$$

$$= z \frac{\partial A_2}{\partial x} - y \frac{\partial A_3}{\partial x} - z \frac{\partial A_1}{\partial y} + x \frac{\partial A_3}{\partial y}$$

$$+ y \frac{\partial A_1}{\partial z} - x \frac{\partial A_2}{\partial z}$$

$$= x \left(\frac{\partial A_3}{\partial y} - \frac{\partial A_2}{\partial z} \right) - y \left(\frac{\partial A_3}{\partial x} - \frac{\partial A_1}{\partial z} \right)$$

$$+ z \left(\frac{\partial A_2}{\partial x} - \frac{\partial A_1}{\partial y} \right)$$

$$= (\underline{x}\underline{i} + \underline{y}\underline{j} + \underline{z}\underline{k}) \cdot \left[\left(\frac{\partial A_3}{\partial y} - \frac{\partial A_2}{\partial z} \right) \underline{i} - \left(\frac{\partial A_3}{\partial x} - \frac{\partial A_1}{\partial z} \right) \underline{j} + \left(\frac{\partial A_2}{\partial x} - \frac{\partial A_1}{\partial y} \right) \underline{k} \right]$$

$$= \underline{r} \cdot (\underline{\nabla} \times \underline{A})$$

$$= \underline{r} \cdot \underline{0} \quad [\because \underline{\nabla} \times \underline{A} = \underline{0}]$$

$$= 0$$

Ans.

□ show that $\underline{\nabla} \cdot (\underline{\nabla} \times \underline{A}) = 0$

Solution:

$$\underline{\nabla} \cdot (\underline{\nabla} \times \underline{A}) = \underline{\nabla} \cdot \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_1 & A_2 & A_3 \end{vmatrix}$$

$$= \underline{\nabla} \cdot \left[\left(\frac{\partial A_3}{\partial y} - \frac{\partial A_2}{\partial z} \right) \underline{i} - \left(\frac{\partial A_3}{\partial x} - \frac{\partial A_1}{\partial z} \right) \underline{j} + \left(\frac{\partial A_2}{\partial x} - \frac{\partial A_1}{\partial y} \right) \underline{k} \right]$$

$$= \frac{\partial}{\partial x} \left(\frac{\partial A_3}{\partial y} - \frac{\partial A_2}{\partial z} \right) - \frac{\partial}{\partial y} \left(\frac{\partial A_3}{\partial x} - \frac{\partial A_1}{\partial z} \right) + \frac{\partial}{\partial z} \left(\frac{\partial A_2}{\partial x} - \frac{\partial A_1}{\partial y} \right)$$

$$= \frac{\partial^2 A_3}{\partial x \partial y} - \frac{\partial^2 A_2}{\partial x \partial z} - \frac{\partial^2 A_3}{\partial y \partial x} + \frac{\partial^2 A_1}{\partial y \partial z} + \frac{\partial^2 A_2}{\partial z \partial x} - \frac{\partial^2 A_1}{\partial z \partial y}$$

$$= 0 \text{ Ans.}$$

Q If $\underline{v} = \underline{\omega} \times \underline{r}$ then prove that
 $\underline{\omega} = \frac{1}{2} \text{curl } \underline{v}$, where $\underline{\omega}$ is a constant vector.

Solution: $\text{Curl } \underline{v}$

$$= \underline{\nabla} \times \underline{v}$$

$$= \underline{\nabla} \times (\underline{\omega} \times \underline{r})$$

$$= \underline{\nabla} \times \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ \omega_1 & \omega_2 & \omega_3 \\ x & y & z \end{vmatrix}$$

$$= \underline{\nabla} \times [(\omega_2 z - \omega_3 y) \underline{i} - (\omega_1 z - \omega_3 x) \underline{j} + (\omega_1 y - \omega_2 x) \underline{k}]$$

$$= \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \omega_2 z - \omega_3 y & -\omega_1 z + \omega_3 x & \omega_1 y - \omega_2 x \end{vmatrix}$$

$$= \underline{i} [\omega_1 + \omega_1] - \underline{j} [-\omega_2 - \omega_2] + \underline{k} [\omega_3 + \omega_3]$$

$$= 2 (\omega_1 \underline{i} + \omega_2 \underline{j} + \omega_3 \underline{k})$$

$$= 2 \underline{\omega}$$

$$\Rightarrow \underline{\omega} = \frac{1}{2} \text{curl } \underline{v} \quad (\text{Proved})$$